1. **Objective**
   - Install Docker Desktop on Windows or macOS.
   - Deploy Gitea service using Docker.
   - Verify Gitea Large File Storage (LFS) support.
   - Create a repository and upload a large file (>1GB).
   - (Optional) Backup and restore Gitea data.
   - (Optional) Install Gitea without Docker.
   - Record key steps with screenshots.
2. **Environment**
   - OS: Windows
   - Tools: Docker Desktop, WSL2 (Windows), Terminal, Browser
   - Software: Gitea (Docker image)
3. **Procedures and Results**

3.1 Install Docker Desktop

1) Install WSL2:

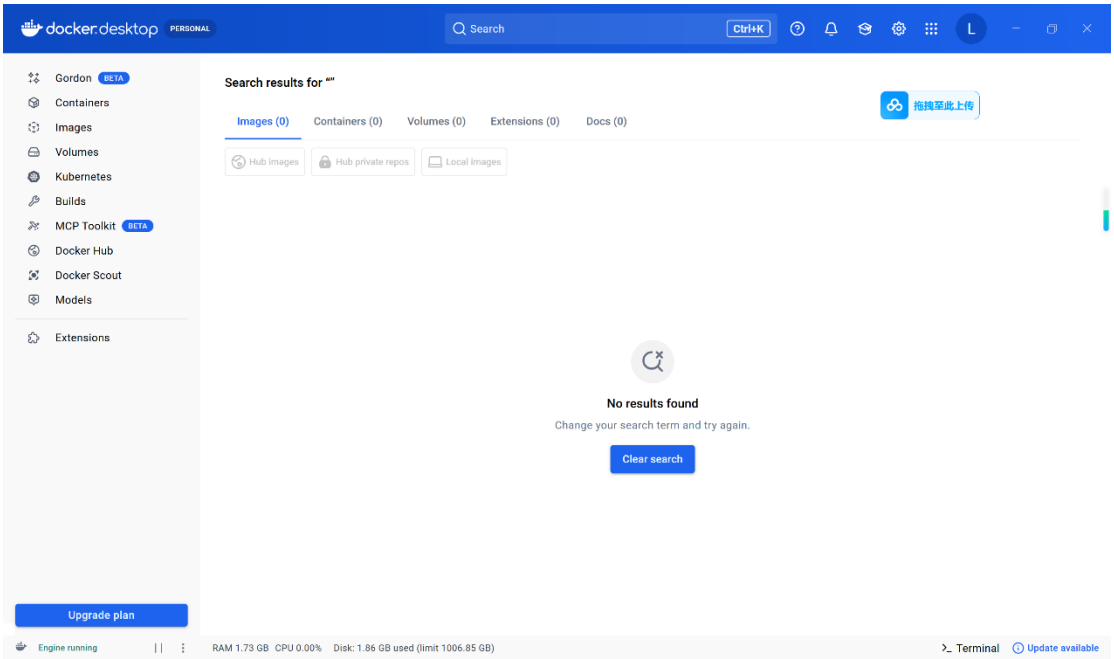
Open PowerShell as administrator and run:

wsl –install

2) Restart the computer.

3) Download and install Docker Desktop, then register and log in.

```
PS C:\Users\36237> docker --version
Docker version 29.3.1, build c2be9cc
```



3.2 Install Gitea via Docker

1) Pull Gitea Docker image:

   docker pull gitea/gitea:latest

2) Create and start Gitea container:

docker run -d --name=gitea -p 3000:3000 -p 222:22 -v gitea_data:/data gitea/gitea:latest

3) Access Gitea Web UI at: https://about.gitea.com/
4) Complete initial configuration (database, admin account).

```
PS C:\Users\36237> docker pull gitea/gitea:latest
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct and if th
e daemon is running: open //./pipe/dockerDesktopLinuxEngine: The system cannot find the file specified.
PS C:\Users\36237> docker pull gitea/gitea:latest
latest: Pulling from gitea/gitea
63ceb4d62a0b: Pull complete
5db3f65dd9c5: Pull complete
d49a2dee86fb: Pull complete
cc4398cab299: Pull complete
0db3a2dc39b9: Pull complete
a0be84997a4f: Pull complete
edafa89f3509: Download complete
Digest: sha256:f846d26a4fc389c5806a580a765e00bfdd1fd181e6f2060da98ea2669d914472
Status: Downloaded newer image for gitea/gitea:latest
docker.io/gitea/gitea:latest
PS C:\Users\36237> docker run -d --name=gitea -p 3000:3000 -p 222:22 -v gitea_data:/data gitea/gitea:latest
2140cf7a69ccbbaadeaf95a57bed3146fef1d0f95a96c83b5a0e33621dd2b0b7
PS C:\Users\36237> ^C
PS C:\Users\36237> docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED          STATUS          PORTS
                                                                                 NAMES
2140cf7a69cc   gitea/gitea:latest   "/usr/bin/entrypoint…"   26 seconds ago   Up 24 seconds   0.0.0.0:3000->3000/tcp, [:
:]:3000->3000/tcp, 0.0.0.0:222->22/tcp, [::]:222->22/tcp   gitea
```

3.3 Verify Gitea LFS Support

1) Go to Gitea administrator settings or repository settings.
2) Check that Git LFS is enabled by default.
3) View documentation: Gitea officially supports Git LFS.
4) Test by attempting to upload a large file; Gitea processes it via LFS.

Conclusion: Gitea supports LFS.

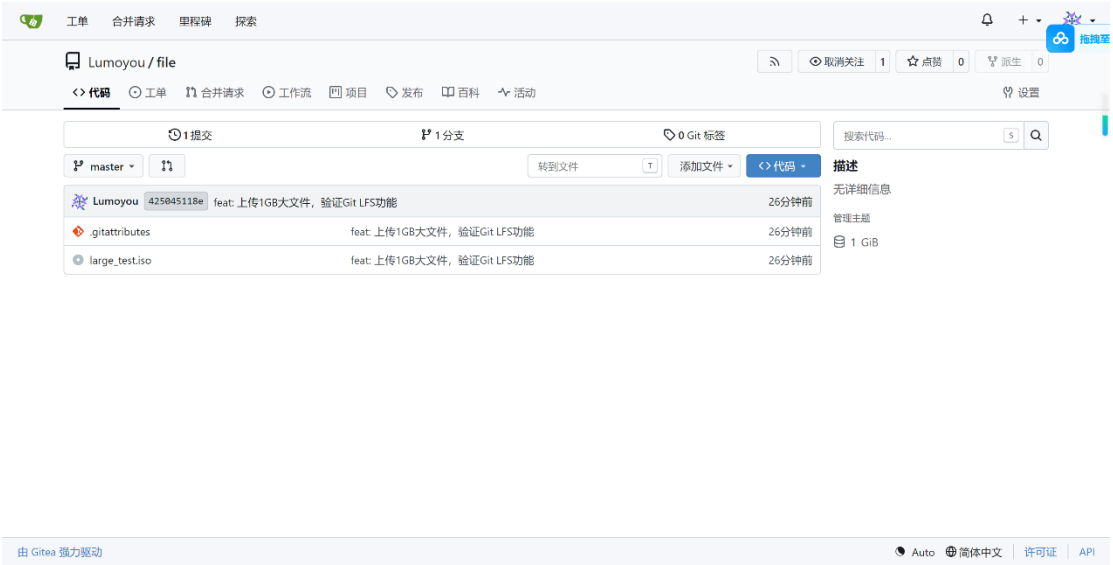
```
PS C:\Users\36237> git lfs install
Git LFS initialized.
```

3.4 Create Repository with Large File (>1GB)

1) Create a new repository in Gitea.
2) Initialize locally and enable LFS
3) Generate a 1GB test file
4) Commit and push to Gitea

```
PS C:\Users\36237> cd D:\gitea-1g-test
PS D:\gitea-1g-test> git init
Initialized empty Git repository in D:/gitea-1g-test/.git/
PS D:\gitea-1g-test> git lfs track "*.iso"
Tracking "*.iso"
PS D:\gitea-1g-test> fsutil file createnew large_1g_test.iso 1073741824
已创建文件 D:\gitea-1g-test\large_1g_test.iso
PS D:\gitea-1g-test> git add .gitattributes large_1g_test.iso
PS D:\gitea-1g-test> git commit -m "feat: 上传1GB大文件，验证Git LFS功能"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got '36237@Yezo.(none)')
PS D:\gitea-1g-test> git config --global user.email "15951915870@163.com"
PS D:\gitea-1g-test>  git commit -m "feat: 上传1GB大文件，验证Git LFS功能"
[master (root-commit) d649176] feat: 上传1GB大文件，验证Git LFS功能
 2 files changed, 4 insertions(+)
 create mode 100644 .gitattributes
 create mode 100644 large_1g_test.iso
```

```
PS D:\gitea-large-file-repo> git remote set-url origin https://gitea.com/Lumoyou/file.git
PS D:\gitea-large-file-repo> git push -u origin master
Locking support detected on remote "origin". Consider enabling it with:
  $ git config lfs.https://gitea.com/Lumoyou/file.git/info/lfs.locksverify true
Uploading LFS objects: 100% (1/1), 1.1 GB | 0 B/s, done.
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 18 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 467 bytes | 467.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: . Processing 1 references
remote: Processed 1 references in total
To https://gitea.com/Lumoyou/file.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
PS D:\gitea-large-file-repo>
PS D:\gitea-large-file-repo>
```



4. Conclusion

In this lab, I successfully installed Docker Desktop, deployed Gitea using Docker, verified LFS functionality, and uploaded a large file greater than 1GB. Optional backup/restore and non-Docker installation were also completed. The experiment shows Gitea is lightweight, easy to deploy via Docker, and fully supports large file storage with LFS.